

Errata and minor updates

Physical Geodesy: A Theoretical Introduction

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1. Page 3, lines 8 to 9:
Change “as attraction, of m ” to “as attraction of m ”
2. Page 5. Fig. 1.3 caption:
Change “acceleration” to “attraction”
3. Page 16, lines 2,3,5, 6 and 7:
Change “ $\hat{e}_h$ ” to “ $\hat{e}_H$ ”
4. Page 16, line 7:
Change “ $\partial W/\partial h$ ” to “ $\partial W/\partial H$ ”
5. Page 16, eqn (1.48):

$$g = -\frac{\partial W}{\partial H}.$$

6. Page 16, eqn (1.49):

$$dH = -\frac{dW}{g}. \quad (1)$$

7. Page 16, title of Sect. 1.3.2:
Change to “Geoid; Dynamic and Orthometric heights”
8. Page 26. Below eqn (1.85) and below eqn (1.88):
Change “disc” to “disk”
9. Page 32, 4th line of Sect. 1.5.2:
Change “disc” to “disk”
10. Page 37. Line 7 from bottom:
Change “A function satisfying” to “A function with continuous second order partial derivatives satisfying”

11. Page 40, below eqn. (1.151):

$$(\Delta x^2 + \Delta z^2)^{1/2} = [1 + (dz/dx)^2]^{1/2} \Delta x$$

12. Page 44. Line 11:

Change “Fig. 1.17b.” to “Fig. 1.17b, where the curve is an infinitesimal fraction.”

13. Page 61, add the following paragraph after the last line:

Readers are suggested to study the uniqueness of the solutions of the boundary value problems for an internal domain, and furthermore, to show that the surface density μ in (2.55) is unique based on the uniqueness of the solutions of the first-type boundary value problems for both an external domain and an internal domain (suggestion: assume V is known on S , and prove that C does not contribute to $(\partial U / \partial n)_i$). Readers are also suggested to compare the expression of μ in (2.55) with (1.130) by setting $\hat{e}_t = \hat{e}_n$.

14. Page 89. eqn (3.45):

$$\delta = \lim_{i \rightarrow \infty} \left| \frac{C_i}{C_{i+2}} \right| = \lim_{i \rightarrow \infty} \left| \frac{(i+2)(i+1)}{i(i+1) - \alpha} \right| = 1.$$

15. Page 91, eqn (3.58):

$$y_1^0(x) = \sum_{i=1}^{\infty} \frac{1}{2i} x^{2i} \quad \text{and} \quad y_2^0(x) = \sum_{i=0}^{\infty} \frac{1}{2i+1} x^{2i+1}$$

16. Page 91, add to the end of the paragraph after (3.58):

The factor M in (3.57) is dropped, since it does not affect the convergence.

17. Page 91, eqn (3.60):

$$\left\{ \begin{array}{l} \frac{1}{2} [\ln(1+x) + \ln(1-x)] = - \sum_{i=1}^{\infty} \frac{x^{2i}}{2i}, \\ \frac{1}{2} [\ln(1+x) - \ln(1-x)] = \sum_{i=0}^{\infty} \frac{x^{2i+1}}{2i+1}. \end{array} \right.$$

18. Page 105, eqn (3.132):

$$P_n^k(\pm 1) = 0, \quad k \neq 0;$$

19. Page 105, eqn (3.134):

Replace the last “.” by “,”

20. Page 107, eqn (3.138):

First line: Change “ $\frac{n!}{(n-k)!}$ ” to “ $\frac{n!}{(i-k)!}$ ”.

Fifth line: Change “ $(x-1)$ ” to “ $(x-1)^n$ ”.

Sixth line: Change “ $(x+1)$ ” to “ $(x+1)^n$ ”.

21. Page 131, eqn (3.270):
Replace “ $k = 1$ ” by “ $k = 0$ ” under the summation sign.
22. Page 136, in equation (3.285):
Change “ C_1^0 ” to “ C_1 ”
23. Page 142, eqn (3.314):

$$V = \frac{GM}{r} \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n \sum_{k=0}^n (\bar{C}_n^k \cos k\lambda + \bar{S}_n^k \sin k\lambda) \bar{P}_n^k(\cos \theta),$$

Remark: Including the degree 1 normalized coefficients is a better choice.

24. Page 144, eqn (3.321):
Replace “ $k = 1$ ” by “ $k = 0$ ” under the summation sign.
25. Page 144, eqn (3.322):
Replace “ $k = 1$ ” by “ $k = 0$ ” under the summation sign.
26. Page 145, eqn (3.327) should be:
Replace “ $k = 1$ ” by “ $k = 0$ ” under the summation sign.
27. Page 145, in the title of Sect. 3.6:
Replace “functions’ by “function”
28. Page 145, left hand side of the second line of eqn (3.329):
Replace “ a_k ” by “ b_k ”
29. Page 147, in eqn (3.335):
Replace “ 2π ” by “ 4π ”
30. Page 153, below eqn (3.370):
Change “Here we ... be written as” to “On the equator, with $\theta = \pi/2$, this formula can be written as”
31. Page 159, last sentence of the Abstract:
Change “called Geodetic Reference System (GRS)” to “which are essential for a Geodetic Reference System (GRS)”
32. Page 188, paragraph below eqn (4.135):
Change “Borkowski (1987)” to “Borkowski (1989)”
33. Page 199, line 7 from bottom:
Change “which are referred to as Geodetic Reference System” to “which are essential for a Geodetic Reference System”
34. Page 200, the line above eqn (4.180):
Change “are the Geodetic Reference System GRS80” to “are those of the Geodetic Reference System 1980 (GRS80)”

35. Page 201. End of the line above eqn (4.195):

Add “,”

36. Page 202, line 7 from bottom, the reference should be replaced by:

Borkowski, K.M. (1989). Accurate algorithms to transform geocentric to geodetic coordinates. Bulletin Géodésique, 63, 50-56.

37. Page 204, the first line of the paragraph above the Section title 5.1.2:

Change “defers” to “differs”

38. Page 209, eqn (5.20):

$$\left(\frac{\partial U}{\partial n_{\Sigma}} \right)_{\Sigma} = \left(\frac{\partial U}{\partial n_S} \right)_S + N \left(\frac{\partial^2 U}{\partial n_S^2} \right)_S = \left(\frac{\partial U}{\partial n_S} \right)_S - N \left(\frac{\partial \gamma}{\partial n_S} \right)_S .$$

39. Page 215, eqn (5.45), last term:

Change “ $+x \cos \psi$ ” to “ $-x \cos \psi$ ”.

40. Page 246, Title of Sect. 5.5.2:

Change “Degree Power Spectrum” to “Degree Variance Power Spectrum”

41. Page 247, eqn (5.172):

$$V_n = \frac{GM}{R} \left(\frac{R}{r} \right)^{n+1} \sum_{k=0}^n \left[\frac{2(2n+1)}{1+\delta_k} \frac{(n-k)!}{(n+k)!} \right]^{1/2} \\ \times (\bar{C}_n^k \cos k\lambda + \bar{S}_n^k \sin k\lambda) P_n^k(\cos \theta) ,$$

42. Page 247, eqn (5.174):

$$\frac{1}{4\pi} \int_{\omega} V^2 d\omega = \left(\frac{GM}{R} \right)^2 \sum_{n=0}^{n_{\max}} \left(\frac{R}{r} \right)^{2n+2} \sum_{k=0}^n \left[\left(\bar{C}_n^k \right)^2 + \left(\bar{S}_n^k \right)^2 \right] \\ = \sum_{n=0}^{n_{\max}} \left(\frac{1}{4\pi} \int_{\omega} (V_n)^2 d\omega \right) .$$

43. Page 248, Title of Fig. 5.10:

Change “Degree power spectrum variance” to “Degree variance power spectrum”

44. Page 249, Title of Sect. 5.5.3:

Change “Degree Power Spectrum” to “Degree Variance Power Spectrum”

45. Page 256, line 4 from bottom:

Delete the sentence “The correction $\Delta_1 g + \Delta_2 g$ is referred to as incomplete Bouguer correction”

46. Page 256, Last line:

Change “Bouguer correction and anomaly” to “Bouguer anomaly”

47. Page 257, lines 2-3 from bottom:

Change “Evidently, $l^2 = r^2 + z^2$, $\sin \beta = z/(r^2 + z^2)^{1/2}$ and $dv = r d\alpha dr dz$ ” to
 “Evidently, $\sin \beta = z/l$, $l = (r^2 + z^2)^{1/2}$ and $dv = r d\alpha dr dz$.”

48. Page 258, eqn (6.13):

$$\Delta_3 g = G\rho \int_0^{2\pi} d\alpha \int_0^a dr \int_0^{\Delta H} \frac{zr}{(r^2 + z^2)^{3/2}} dz,$$

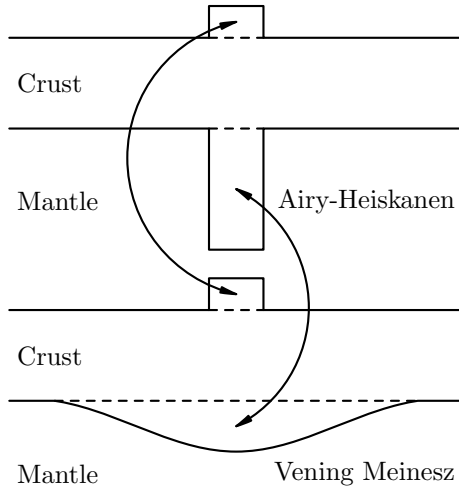
49. Page 258:

Change the sentence above (6.15) to: “The gravity anomaly computed according to”

50. Page 263, eqn (6.27):

$$\Delta\rho = \rho - \rho' = -\frac{(\rho - \rho_w)P}{D - P}.$$

51. Page 265, Fig. 6.7 (It makes me feel embarrassed to have the one in the print! I went to replace it before the submission of the final version to Springer, but I have mistakenly replaced it in a different draft):



52. Page 301, Line 5:

Change “~ 10 m for the isostatic anomaly.” to “~ 10 m for the isostatic anomaly (Heiskanen and Moritz, 1967).”

53. Page 313, eqn (6.223):

$$f_T(T, H', \psi) = \frac{R^2}{2} \left\{ (x + 3 \cos \psi)(1 + x^2 - 2x \cos \psi)^{1/2} + (3 \cos^2 \psi - 1) \right. \\ \left. \times \ln \left[(1 + x^2 - 2x \cos \psi)^{1/2} + x - \cos \psi \right] \right\}^{1-T/R}_{1-(T+H')/R}.$$

54. Page 319:
Add the following into the references:
Heiskanen, W.A., & Moritz, H. (1967). *Physical geodesy*. W.H. Freeman and Company.
55. Page 319, add to further reading:
Szwilius, W., Ebbing, J., & Holzrichter, N. (2016). Importance of far-field topographic and isostatic corrections for regional density modelling. *Geophysical Journal International*, 207, 274–287. <https://doi.org/10.1093/gji/ggw270>
56. Page 347, 3rd line above eqn (7.116):
Change
be readily verified using (7.24) that ... implies “ $r\Delta G$ is harmonic.”
to
be readily verified using (5.27) and (7.24) that $r\partial T/\partial r$ and $r\Delta G$ are harmonic.
57. Page 393 line 3 and page 493 line 7 from bottom (So embarrassed to misspell a colleague’s name):
Change “Bgherbandi” to “Bagherbandi”.
The same correction applies to Page 493, line 7 from the bottom.
58. Page 398, line 26 (or line 9 of the third paragraph):
Change “If not, the difference between the differences” to “If not, the sum of the two differences”
59. Page 410, line 6:
Change “ $n_{\max}(n_{\max} + 1)$ ” to “ $(n_{\max} + 1)^2$ ”
60. Page 415, line 2 of the last paragraph:
Add “, although the decrease may be oscillatory” after “ β decreases with n ”
61. Page 417, line 3 from bottom:
Change “ $\int_{-\sigma}^{\sigma} g(x)dx = 99.73\%$ ” to “ $\int_{-3\sigma}^{3\sigma} g(x)dx = 99.73\%$ ”
62. Page 419:
Add a “,” after (8.75) just above eqn (8.78).
63. Page 423: Eqn (8.98) should be

$$\sum_{i=1}^N \sum_{j=1}^{2N} W_{ij} \left[\sum_{n=2}^{N-1} \sum_{k=0}^n (\alpha_{ijnk} \bar{c}_n^k + \beta_{ijnk} \bar{s}_n^k) - \Delta \bar{g}_{ij} \right]^2 = \min$$

64. Page 424: Change the paragraph starting the 3rd line to
“The second problem is the resolvability. In this approach, the coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ can be reliably solved only up to degree and order $N - 1$ (Colombo,

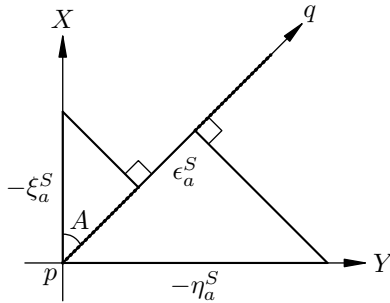
1981). This could be demonstrated by examining the problem on the equator where $\theta = \pi/2$ and (8.32) degenerates to the Fourier series

$$\Delta g(\lambda) = \sum_{k=0}^{n_{\max}} (a_k \cos k\lambda + b_k \sin k\lambda). \quad (8.99)$$

Hence, there are $2n_{\max}+1$ coefficients to be determined. However, there are $2N$ values of Δg on the equator. The total number of coefficients to be determined cannot be more than the total number of data. As a result, the coefficients can only be solved up to order $n_{\max} = N - 1$ at most. With $n_{\max} = N - 1$ as chosen in (8.98), the total number of coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ is N^2 , while the total number of data $\Delta \bar{g}_{ij}$ is $2N^2$."

65. Page 438, second line below 8.5.1:
Change "base" to "based"

66. Page 439, Fig. 8.13:



67. Page 450. eqn (9.14):
Change " $\frac{d}{dr}$ " to " $\frac{d}{dr}$ "
68. Page 452. eqn (9.19):
Change " $Y_n^k(\theta, \lambda)$ " to " $Y_n(\theta, \lambda)$ "
69. Page 461, the line above (9.66):
Change "We now assume" to "We now make the finer approximations"
70. Page 461, the line below (9.66):
Change "this assumption" to "these approximations"
71. Page 487. Add the following by the beginning of the line below eqn (A.57):
The substitution of (A.54) into this equation yields an equation for B , which is referred to as the latitude equation in the literature.